Paleomagnetism and K–Ar isochron dates of Early Jurassic basaltic flows and dikes of Atlantic Canada

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K–Ar isochron dates of 193 ± 2, 189 ± 3, and 191 ± 2 Ma are reported for the Shelburne dike of Nova Scotia, the Avalon dike of Newfoundland, and the Caraquet dike of New Brunswick, respectively. These major tholeiitic dikes agree in radiometric date with the North Mountain Basalt of Nova Scotia and with other tholeiitic flows of the Newark Supergroup that have been paleontologically assigned to the Hettangian. However, further work is needed to resolve the discrepancy between the Triassic–Jurassic boundary date of 193 ± 6 Ma suggested by the Newark Supergroup flows and the boundary dates of 204–213 Ma adopted in recent time scales.

The North Mountain Basalt flows at nine sites on the north shore of the Bay of Fundy are shown to have likely magnetized during initial cooling, yielding an Hettangian virtual paleopole at 66°N, 72°E (dp = 9°, dm = 14°). New paleomagnetic results are also reported for the Caraquet and Avalon dikes.

The virtual paleopoles from Hettangian flows and dikes of Atlantic Canada lie on average ~11° north of the virtual paleopoles from Hettangian flows of the northeastern United States. This discrepancy is likely due to insufficient averaging out of paleosecular variation and demonstrates that caution is needed in paleomagnetically correlating among the Early Jurassic flows and intrusions of eastern North America.

Le dyke de Shelburne de la Nouvelle-Écosse, le dyke d'Avalon de Terre-Neuve et le dyke de Caraquet du Nouveau-Brunswick ont fourni des dates isochrones K–Ar de 193 ± 2, 189 ± 3 et 191 ± 2 Ma, respectivement. Les âges radiométriques de ces dykes tholéiitiques majeurs s'accordent avec les données paléontologiques qui assignent à l'Hettangien le basalte de North Mountain de la Nouvelle-Écosse et autres coulées tholéiitiques du Supergroupe de Newark. Cependant, une étude plus détaillée est nécessaire pour comprendre l'écart la date de 193 ± 6 Ma de la limite Trias–Jurassique suggérée par les coulées du Supergroupe de Newark et les dates limites de 204 à 213 Ma adoptées dans les échelles de temps récentes.

Les coulées de basalte de North Mountain, échantillonnées à neuf sites localisés sur la rive nord de la baie de Fundy, apparissent avoir été semblablement aimantées durant le refroidissement initial, et elles fournissent un paléopôle virtuel hettangien à 66°N, 72°E (dp = 9°, dm = 14°). De nouveaux résultats paléomagnétiques sont aussi présentés pour les dykes de Caraquet et d'Avalon.

Les pôles virtuels des coulées et des dykes hettangiens des provinces de l'Atlantique du Canada sont orientés en moyenne à ~11° nord des paléopôles virtuels des coulées hettangiennes du nord-est des États-Unis. Cette différence est probablement due à une moyenne insuffisante de la variation paléoséculaire et démontre qu'il faut user de prudence en tentant d'établir des corrélations entre les données paléomagnétiques des coulées et des intrusions du Jurassique précoce de l'est de l'Amérique du Nord.

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Introduction

In the Late Triassic – Early Jurassic, rift basins formed along North America's eastern margin and filled with continental sediments of the Newark Supergroup. Among these sediments are tholeiitic basalt flows extruded during the Early Jurassic just prior to initiation of Atlantic sea-floor spreading (Van Houten 1977; Manspeizer *et al.* 1978). In Atlantic Canada, these tholeiites of the Newark Supergroup are represented by the North Mountain Basalt flows of the Fundy Basin, to which three major tholeiitic dikes may be related: the Shelburne dike of Nova Scotia, the Avalon dike of Newfoundland, and the Caraquet dike of New Brunswick (Fig. 1). In this paper, we test the correlation of these three major dikes with the North Mountain Basalt flows using K–Ar isochron dating, and we compare the virtual paleopole positions yielded by these basalts of Atlantic Canada and by their correlatives in the northeastern United States.

The North Mountain Basalt flows of the Fundy Basin have been paleontologically assigned (Olsen *et al.* 1982) to the earliest Jurassic (Hettangian Age) along with most of the other basalt flows of the Newark Supergroup (Fig. 1). This age assignment is supported by palynology and by the recent discovery in sediments directly above the North Mountain Basalt of abundant tetrapod fossils thought to represent survivors of a terminal Triassic extinction event (Olsen *et al.* 1987). It has been suggested that the Shelburne dike was a feeder to the North Mountain Basalt (Aumento 1966) and is related to the Avalon dike, with which it is approximately on strike (Papezik *et al.* 1975). We began testing these correlations earlier (Hodych and Hayatsu 1980) comparing preliminary K–Ar isochrons for these two dikes with the K–Ar isochrons for the North Mountain Basalt (Hayatsu 1979); we shall now complete these two dike isochrons. The Caraquet dike may also be related to the Shelburne dike, which it roughly parallels. However, Seguin *et al.* (1981) suggested from paleomagnetic data that there may have been a second intrusion along the

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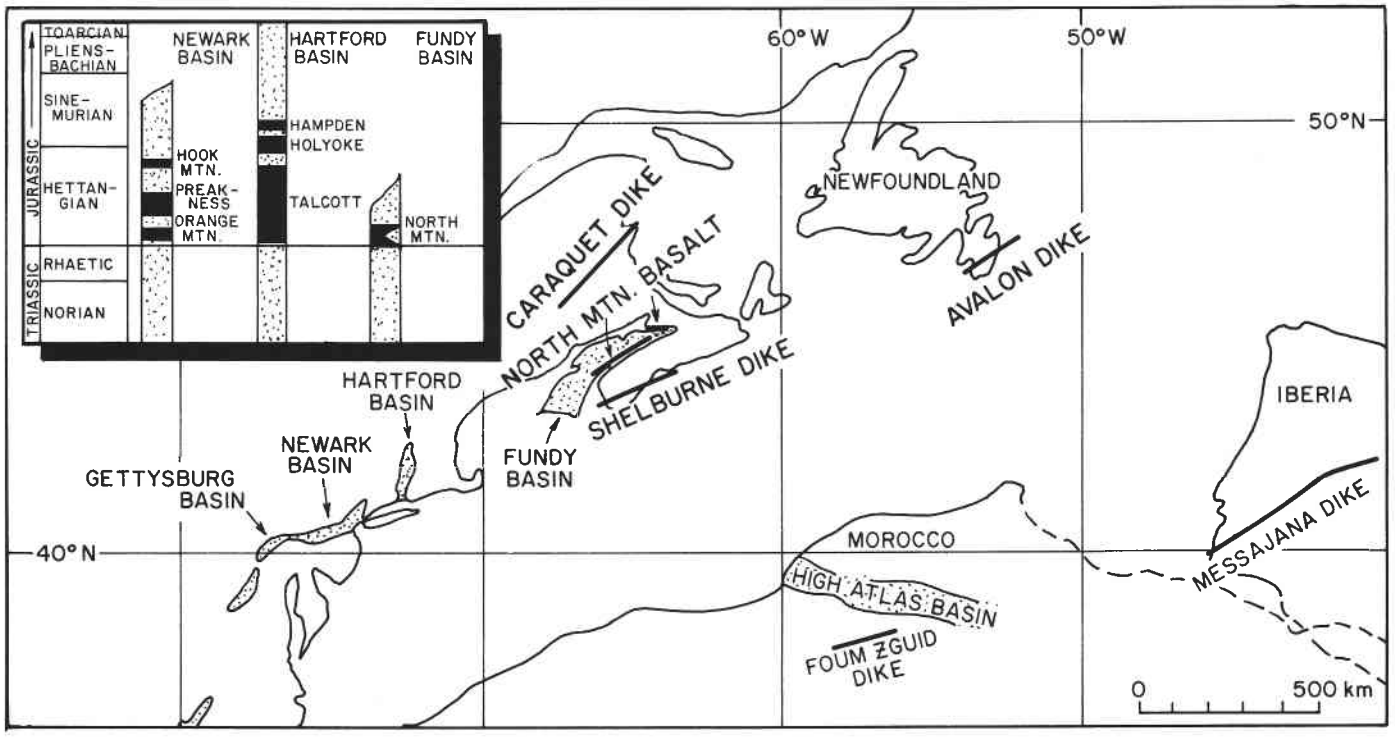


FIG. 1. The Caraquet, Shelburne, and Avalon dolerite dikes of Atlantic Canada with their possible trans-Atlantic equivalents: the Messajana dike and the Foum Zguid dike, shown in predrift configuration (after Sichler *et al.* 1980). Also shown (stippled) are the Late Triassic – Early Jurassic rift basins of eastern North America and Morocco. The inset shows some of the basalt flows (in black) in the basins correlated paleontologically by Olsen *et al.* (1982).

Caraquet dike, perhaps over 20 Ma later than extrusion of the North Mountain Basalt. We test this suggestion with a K–Ar isochron for the Caraquet dike.

The virtual paleopoles yielded by the North Mountain Basalt of the south shore of the Bay of Fundy (Carmichael and Palmer 1968), by the Shelburne dike (Larochelle and Wanless 1966), and by the Caraquet dike (Seguin *et al.* 1981)—all thought to be Hettangian—are considerably more northerly than the virtual paleopoles yielded by the Hettangian flows of the northeastern United States (Fig. 9). To try to resolve this discrepancy, we present the first paleomagnetic results from the North Mountain Basalt of the north shore of the Bay of Fundy and from Avalon dike and Caraquet dike samples yielding approximately the same K–Ar isochron age as the North Mountain Basalt.

The K–Ar isochron dating in this paper was done by Hayatsu; paleomagnetism and most of the interpretation were the responsibility of Hodych.

Sampling sites

Site locations are given in Figs. 2 and 3 and in Table 5. Four to six oriented block samples were collected at each site with the aid of a sun compass (except for the Shelburne dike samples, none of which were oriented).

North Mountain Basalt flows sampled at sites N-1 to N-10 along the north shore of the Bay of Fundy are paleontologically assigned (Olsen *et al.* 1987) to the Hettangian and correlate (Olsen 1981) with the flows of the south shore of the Bay of Fundy. Along the south shore, a thick flow (~190 m) is overlain by thin flows (totaling ~50 m) and a final thick flow (~160 m) (Papezik *et al.* 1988). The north shore is block faulted, making it difficult to determine position in the sequence of flows; hence, the positioning is partly based on $^{87}Sr/^{86}Sr$ ratios by J. D. Greenough (personal communication, 1988). A thick lowermost flow was likely sampled at sites N-4 and N-9 and perhaps at N-10. Thin middle flows may have been sampled at sites N-1, N-2, and N-3. A thick upper flow may have been sampled at sites N-5, N-7, and N-8; a thin overlying flow, at N-6.

The Shelburne dike sample S-A 9 was collected at site S-A in an abandoned quarry. Sample S-A A was collected by W. F. Fahrig at site A of Larochelle and Wanless (1966), which is near our site S-A. We also collected at site S-B in Back Cove in the town of Cherry Hill. Samples S-B 1 and 2 were collected on the west side of Back Cove, whereas samples S-B 3, 4, 5, G, Q, F, and C were collected from the east side of Back Cove. Sample S-B B was collected by W. F. Fahrig at site B of Larochelle and Wanless (1966), which is near our site S-B.

The Avalon dike was inferred from its aeromagnetic lineament (Papezik *et al.* 1975), and the few outcrops known are shown in Fig. 3. Whereas previous sampling (sites A-1 and A-2) was in minor intrusive bodies along the western part of the lineament (Hodych and Hayatsu 1980), sites A-3a, A-3b, and A-3c are along the eastern part of the lineament in coarser grained dolerites, presumably representing the major dike(s) responsible for the lineament. Sites A-3a and A-3b are the most accessible, occurring in small outcrops to the west and east, respectively, of an abandoned railway bed in La Manche Provincial Park. At site A-3c, only large angular boulders rather than outcrop could be sampled (Hodych and Hayatsu 1980).

The Caraquet dike was sampled at site C-1 in a quarry near Pointe à Brideau in Caraquet a few hundred metres to the west

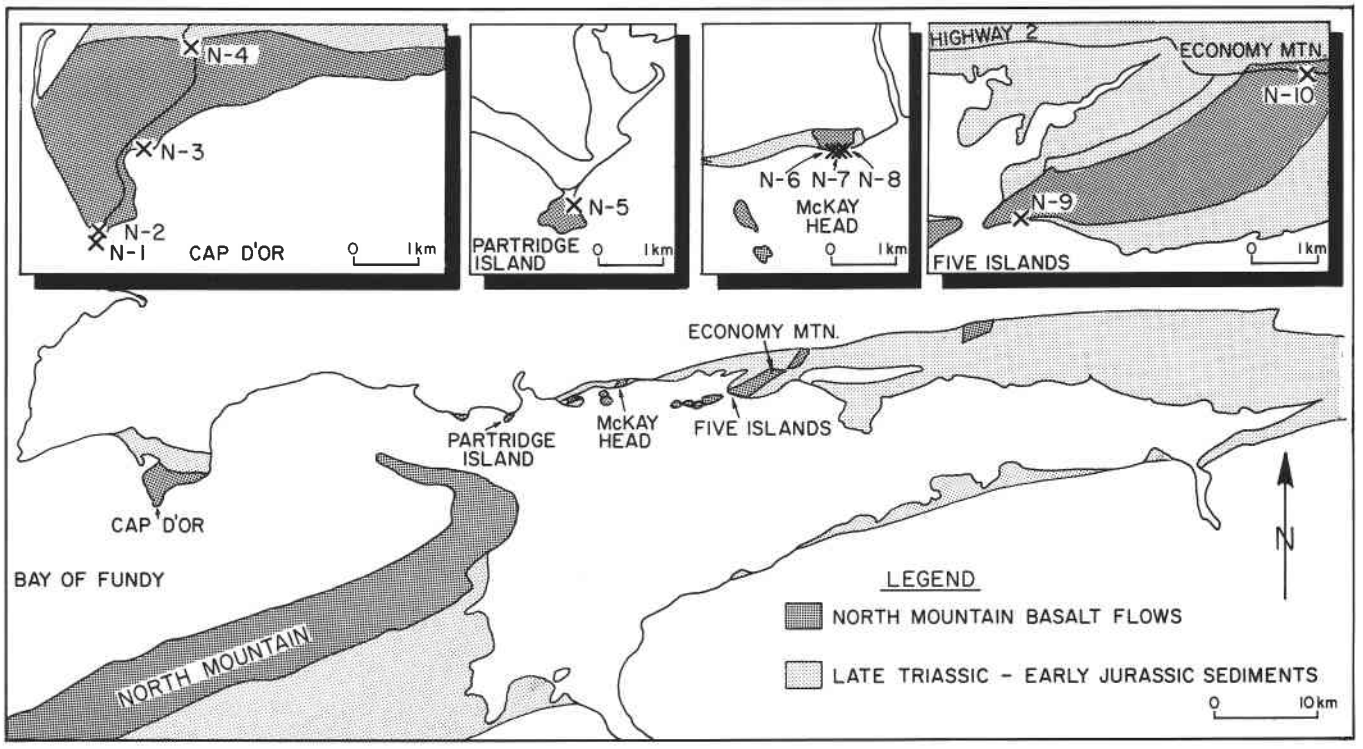


FIG. 2. Sampling sites in the North Mountain Basalt flows of the north shore of the Bay of Fundy. Geology after Donohoe and Wallace (1982) and Bujak and Donohoe (1980).

of site C (no longer accessible) of Seguin *et al.* (1981). We also sampled at site C-2 on the north shore of Caraquet Island.

K–Ar isochron dating

K–Ar isochron dating (Hayatsu and Carmichael 1977) was applied to whole-rock dolerite samples using the same analytical procedure as outlined in Hodych and Hayatsu (1980). The analytical data are presented in Table 1 and plotted in Fig. 4.

For the Shelburne dike, thin sections were examined by J. D. Greenough for all 12 samples used for dating. Most samples are fresh. Samples 1, 2, 9, and A contain about 15% hornblende, which was likely the result of deuteric alteration and is thus not expected to have affected the isochron age. Sample 3 contains about 3% chlorite and actinolite, which may also be deuteric in origin. Sample 4 contains much altered glass and hence is likely the most unreliable of the samples. We prefer to exclude sample 4, in which case the K–Ar isochron selected using least-squares cubic fitting (York 1969) yields an age of 193.2 ± 1.6 (2σ) Ma, and the isochron's intercept on the $^{40}Ar/^{39}Ar$ axis is at 229 ± 24 Ma. (If all 12 samples are used, the isochron yields an age of 195.8 ± 3.4 (2σ) Ma, and the isochron's intercept on the $^{40}Ar/^{39}Ar$ axis is at 205 ± 53 Ma.) The conventional K–Ar date from Larochelle and Wanless (1966) is in agreement but has a very large analytical error (Table 3).

For the Avalon dike, the eight point K–Ar isochron selected using least-squares cubic fitting yields an age of 188.9 ± 3.0 (2σ) Ma, and its intercept on the $^{40}Ar/^{36}Ar$ axis is at 267 ± 32 Ma. The isochron depends heavily upon samples 3 and 4 from site A-3c, which are not ideal samples, being from large angular boulders rather than outcrop and perhaps being slightly weathered. Nevertheless, all the samples—even those from the fresh outcrops at sites A-3a and A-3b—fit the 190 Ma isochron better than they would the 201 Ma isochron obtained (Hodych and Hayatsu 1980) for the minor intrusions at sites A-1 and A-2.

For the Caraquet dike, the seven point K–Ar isochron selected using least-squares cubic fitting yields an age of 190.6 ± 2.4 (2σ) ma, and its intercept on the $^{40}Ar/^{39}Ar$ axis is at 251 ± 28 Ma. The average of two conventional K–Ar dates by Wanless *et al.* (1972) does not differ significantly from this date but has very poor precision (Table 3).

Interpretation of the radiometric dates

Dating of Hettangian flows and intrusions of eastern North America

Most flows of the Newark Supergroup (Fig. 1) were assigned to the Hettangian by Olsen *et al.* (1982) using paleontological evidence, which will be discussed below. All radiometric dates for Newark Supergroup flows paleontologically assigned to the Hettangian are listed in Table 2. (Throughout this paper, unpublished dates or dates that have only been published in abstracts or at second hand are excluded; all dates have been corrected to conform to the constants of Table 1 and are given with estimates of error calculated as explained in the footnotes to Table 2). No flow's date differs significantly from the average radiometric date for all the flows, supporting the paleontological correlation. The six sets of Hettangian flows of Table 2 yield an average date of 190 ± 3 Ma (excluding the two dates with error greater than ±10 Ma).

The Shelburne, Avalon, and Caraquet dikes of Atlantic Canada also yield K–Ar isochron dates near 190 Ma, suggesting that they too are Hettangian in age (with evidence at the western end of the Avalon dike of precursor intrusions about 10 Ma earlier (Hodych and Hayatsu 1980)). This is not unexpected. It has long been suggested that the Shelburne dike may

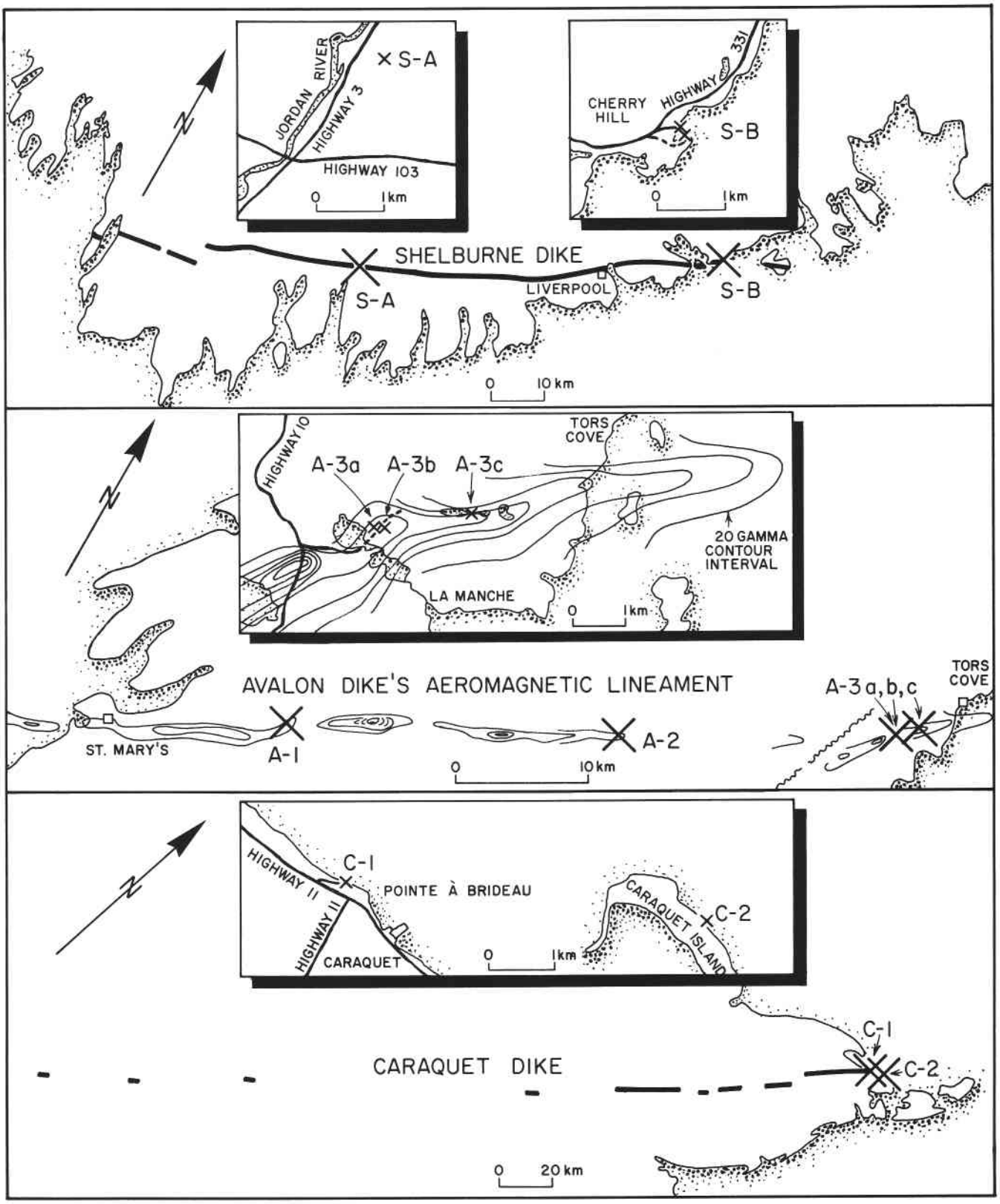


FIG. 3. Sampling sites in the Shelburne, Avalon, and Caraquet dolerite dikes of Atlantic Canada. Geology after Bujak and Donohoe (1980), Hodych and Hayatsu (1980), and Ferguson Fyffe (1985), respectively.

have been a feeder to the North Mountain Basalt flows (Aumento 1966). The aeromagnetic lineament from which the Avalon dike is inferred is approximately on strike with the Shelburne dike, suggesting that they may be related (Papezik *et al.* 1975), and the Caraquet dike very roughly parallels the Shelburne and Avalon dikes.

Of the dikes and sills cutting the Newark Supergroup, only the Fairhaven dike and perhaps the Palisades sill can presently

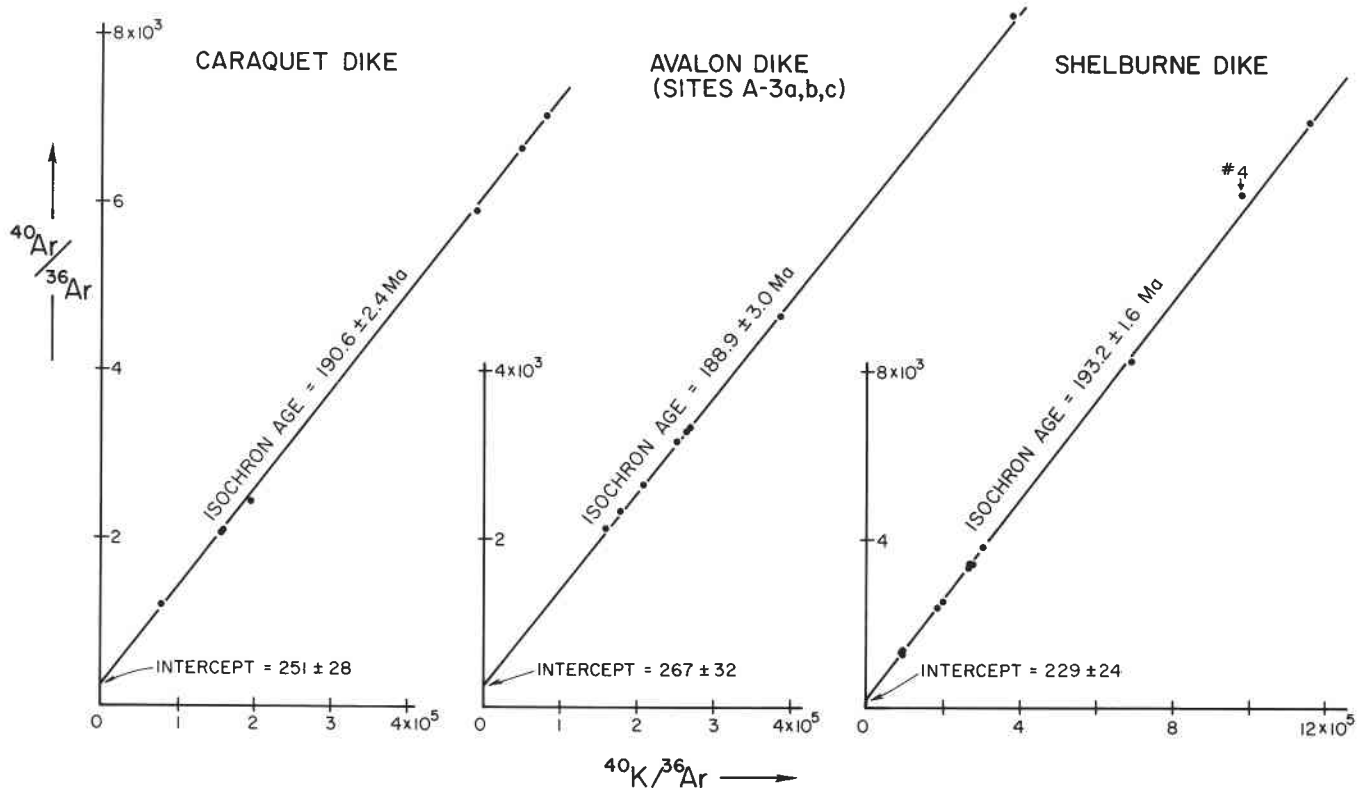


FIG. 4. K–Ar isochrons for the Caraquet, Avalon, and Shelburne dolerite dikes of Atlantic Canada.

TABLE 1. K–Ar analytical data

Site and sample designation	K (%)	$^{40}Ar_{rad}$ ($\times 10^{-6}$ cm^3/g)	Conventional age (Ma)	$^{40}Ar_{total}$ ($\times 10^{-6}$ cm^3/g)	$^{36}Ar_{initial}$ ($\times 10^{-9}$ cm^3/g)	$^{40}K/^{36}Ar$ ($\times 10^5$)	$^{40}Ar/^{36}Ar$
A-3a 4	0.3559	2.772	190.0	3.173	1.356	1.756	2 339
A-3a 5	0.4107	3.166	188.1	3.489	1.096	2.508	3 185
A-3a 6	0.3693	2.870	189.6	3.329	1.552	1.592	2 144
A-3a D	0.3556	2.720	186.7	3.059	1.149	2.071	2 663
A-3b 4	0.4601	3.511	186.3	3.857	1.172	2.628	3 292
A-3b 6	0.4003	3.035	185.2	3.330	0.999	2.681	3 333
A-3c 3	0.7284	5.518	185.0	5.890	1.260	3.867	4 673
A-3c 4	0.7665	5.943	189.1	6.163	0.746	6.877	8 264
S-A 9	0.5793	4.430	186.7	5.004	1.942	1.996	2 577
S-A A	0.8833	6.845	189.0	7.509	2.247	2.631	3 343
S-B 1	0.6345	4.861	187.0	5.321	1.557	2.737	3 417
S-B 2	0.7085	5.422	186.8	6.183	2.576	1.840	2 401
S-B 3	0.5723	4.399	187.6	5.643	4.213	0.909	1 340
S-B 4	0.5237	4.275	198.6	4.381	0.358	9.787	12 233
S-B 5	0.4523	3.531	190.4	3.825	0.993	3.046	3 853
S-B B	0.6611	5.029	186.0	6.385	4.585	0.964	1 392
S-B G	0.6787	5.360	192.5	5.860	1.694	2.681	3 459
S-B Q	0.6024	4.670	189.1	4.843	0.583	6.915	8 308
S-B F	0.6625	5.252	193.2	5.366	0.384	11.552	13 981
S-B C	0.7908	5.934	183.4	7.631	5.741	0.922	1 329
C-1 2	0.3102	2.4099	189.5	3.192	2.647	0.784	1 207
C-1 3	0.4251	3.2393	186.1	3.779	1.825	1.558	2 070
C-1 5	0.5286	4.0686	187.8	4.733	2.250	1.572	2 104
C-2 8	1.326	9.8632	182.1	11.211	4.560	1.942	2 459
C-2 1	0.3968	3.0573	188.0	3.218	0.544	4.878	5 912
C-2 3	0.4257	3.3121	189.8	3.466	0.520	5.479	6 666
C-2 4	0.4336	3.3802	190.1	3.528	0.501	5.796	7 047

NOTES: The constants used were $\lambda_e = 0.581 \times 10^{-10}$ year^{-1}; $\lambda_\beta = 4.962 \times 10^{-10}$ year^{-1}; atomic $^{40}K/K = 1.167 \times 10^{-4}$.

TABLE 2. Radiometric dating of basaltic flows and intrusions of the Newark Supergroup paleontologically assigned to the Hettangian

Unit	Location	Radiometric dating method[a]	Number of samples	Radiometric date (Ma)[b]	Source
North Mountain Basalt flows	Fundy Basin	K−Ar isochron	7	191±5	Hayatsu (1979)
North Mountain Basalt flows	Fundy Basin	K−Ar	5	229±53[c]	Carmichael and Palmer (1968)
Orange Mountain Basalt flows	Newark Basin	K−Ar, plagioclase	1	191±8	Seidemann *et al.* (1984)
Preakness Basalt flows	Newark Basin	K−Ar, plagioclase	5	194±4	Seidemann *et al.* (1984)
Hook Mountain Basalt flows	Newark Basin	K−Ar, plagioclase	4	185±4	Seidemann *et al.* (1984)
Talcott Basalt flows	Hartford Basin	K−Ar	10	186±3	Seidemann *et al.* (1984)
Talcott Basalt flows	Hartford Basin	$^{40}Ar/^{39}Ar$ plateau	1	187±10	Seidemann *et al.* (1984)
Talcott Basalt flows	Hartford Basin	K−Ar	6	187±8	Seidemann (1988)
Holyoke Basalt flows	Hartford Basin	K−Ar, plagioclase	2	189±6	Seidemann *et al.* (1984)
Holyoke Basalt flows	Hartford Basin	K−Ar	4	188±19[c]	Armstrong and Besancon (1970)
Palisades sill	Newark Basin	$^{40}Ar/^{39}Ar$ plateau	2	193±6[c]	Dallmeyer (1975)
Palisades sill	Newark Basin	K−Ar, biotite	2	194±8[c]	Erickson and Kulp (1961)
Palisades sill	Newark Basin	K−Ar, biotite	1	196±4[c]	Armstrong and Besancon (1970)
Fairhaven dike	Hartford Basin	$^{40}Ar/^{39}Ar$ plateau	1	185±7[c]	Sutter and Smith (1979)
Fairhaven dike	Hartford Basin	K−Ar	2	193±18[c]	Armstrong and Besancon (1970)

[a]Radiometric dating done on whole rocks except where a mineral (plagioclase or biotite) is mentioned.
[b]Errors quoted are $\pm 2\sigma/\sqrt{n}$ or the equivalent (estimating 95% confidence limits if the number of samples, n, is large); σ is the standard deviation.
[c]Dates corrected to conform to constants of Table 2 (Harland *et al.* 1982, Appendix 1).

TABLE 3. Basaltic intrusions possibly related to Hettangian flows of the Newark Supergroup, judging by radiometric dates

Intrusion	Location	Radiometric dating method[a]	Number of samples	Radiometric age (Ma)[b]	Source
Avalon dike (east end)	Newfoundland	K−Ar isochron	8	189±3	This paper
Shelburne dike	Nova Scotia	K−Ar isochron	11	193±2	This paper
Shelburne dike	Nova Scotia	K−Ar	1	201±32[c]	Larochelle and Wanless (1966)
Caraquet dike	New Brunswick	K−Ar isochron	7	191±2	This paper
Caraquet dike	New Brunswick	K−Ar	2	177±30[c]	Wanless *et al.* (1972)
Mt. Carmel sill	Hartford Basin	$^{40}Ar/^{39}Ar$ plateau	1	195±8[c]	Sutter and Smith (1979)
Cross Rock dike	Hartford Basin	$^{40}Ar/^{39}Ar$ plateau	1	185±7[c]	Sutter and Smith (1979)
Buttress dike	Hartford Basin	$^{40}Ar/^{39}Ar$ plateau	1	185±13[c]	Sutter and Smith (1979)
Buttress dike	Hartford Basin	K−Ar	5	218±11[c]	Armstrong and Besancon (1970)
West Rock sill	Hartford Basin	K−Ar	3	201±13[c]	Armstrong and Besancon (1970)
Higganum dike	Connecticut	K−Ar	2	197±18[c]	Armstrong and Besancon (1970)
Gettysburg sill	Gettysburg Basin	$^{40}Ar/^{39}Ar$ plateau	2	181±12[c]	Sutter and Smith (1979)
Frederick dikes	Gettysburg Basin	$^{40}Ar/^{39}Ar$ plateau	2	191±14[c]	Sutter and Smith (1979)

[a]Radiometric dating done on whole rocks except where a mineral (plagioclase or biotite) is mentioned.
[b]Errors quoted are $\pm 2\sigma/\sqrt{n}$ or the equivalent (estimating 95% confidence limits if the number of samples, n, is large).
[c]Dates corrected to conform to constants of Table 2 (Harland *et al.* 1982, Appendix 1).

be paleontologically assigned to the Hettangian. Field relations show (Philpotts and Martello 1986) that the Fairhaven dike fed the Hettangian Talcott flows. The dike's $^{40}Ar/^{39}Ar$ plateau age of 185± 7 Ma (Sutter and Smith 1979) does agree with the dates for the Talcott flows. Magnetic and gravity anomalies suggest that the Palisades sill acted as feeder to the Ladentown flows (Kodama 1983), which are thought to be an outlier of the Hettangian Orange Mountain Basalt flows (Olsen 1980). The sill's radiometric dates do agree with the one date from the Orange Mountain flows (Table 2). These two intrusions and the six sets of Hettangian flows yield an average date that is again 190 ± 3 Ma.

The remaining radiometrically dated dikes and sills cutting the Newark Supergroup (see Sutter and Smith 1979) are not very useful in dating the Hettangian because they seem to cut only the Triassic part of the section but otherwise are not paleontologically constrained in age. Of these intrusions, most have radiometric dates whose error estimates overlap those of 190 ± 3 Ma, the average date of the Hettangian flows and intrusions. Hence, these intrusions may also be Hettangian and are listed in Table 3. The only intrusions cutting the Newark Supergroup that have dates sufficiently different from 190 ± 3 Ma for their error estimates not to overlap seem to be two dikes from the Gettysburg and Deep River basins that yield K−Ar dates of 230 and 249 Ma, respectively (Armstrong and Besancon 1970).

Geochemistry of Hettangian flows and intrusions of eastern North America

Is the geochemistry of the basaltic flows and intrusions considered above consistent with their inferred Hettangian age?

First, consider the flows and intrusions of Table 2. The lowermost sets of basalt flows and the two intrusions can be classed with the high-TiO_2 quartz-normative tholeiites of Weigand and Ragland (1970). This was shown for the Orange Mountain Basalt, the Talcott Basalt, and the Palisades sill by

Puffer and Lechler (1980), for the Fairhaven dike by Philpotts and Martello (1986), and for the North Mountain Basalt by Papezik *et al.* (1988). The magma that produced the lowermost sets of basalt flows could have fractionated to produce the next flows—the Preakness and Holyoke basalts (Puffer and Lechler 1980; Philpotts and Reichenbach 1985). Finally, this fractionated magma could have fractionated further in a near-surface magma chamber to produce the youngest basalt flows—the Hook Mountain and Hampden basalts (extending the argument of Philpotts and Reichenbach (1985) for the Hampden Basalt to the Hook Mountain Basalt, since these two basalts are chemically almost identical, according to Puffer and Lechler (1980)). Hence, all the flows and intrusions of Table 2 may be related geochemically.

As for the intrusions of Table 3, the Shelburne, Avalon, Higganum, and Buttress dolerites can all be classed with the high-TiO_2 quartz-normative tholeiites of Weigand and Ragland (1970). This was shown by Papezik and Barr (1981) for the Shelburne dike, by Papezik and Hodych (1980) for the Avalon dike, and by Philpotts and Martello (1986) for the Higganum and Buttress dikes. The Caraquet dolerite lies between the high-TiO_2 and the low-TiO_2 quartz-normative tholeiites of Weigand and Ragland (1970), according to Greenough and Papezik (1986), and the Gettysburg dolerites include both these tholeiite types, according to R. C. Smith *et al.* (1975). Hence, the intrusions of Table 3 may be related geochemically to the Hettangian basaltic flows and intrusions of Table 2.

Radiometric dating of the Triassic–Jurassic boundary

The radiometrically dated flows and intrusions of the Newark Supergroup that have been paleontologically assigned to the Hettangian (Table 2) yield self-consistent radiometric dates averaging 190 ± 3 Ma. This suggests that the age of the Triassic–Jurassic boundary is 193 ± 6 Ma (assuming that the Hettangian lasted between 3 and 7 Ma, as in the time scales below). This agrees with the radiometric time scale of Van Hinte (1976) but is about 15 Ma lower than in most more recently proposed times scales. For example, the Triassic–Jurassic boundary was placed at 204 ± 4 Ma by Kennedy and Odin (1982), at 208 Ma by Kent and Gradstein (1985), and at 213 Ma by Harland *et al.* (1982). Most of these more recent time scales have been strongly influenced by radiometric dates from the Late Triassic – Early Jurassic of British Columbia used by Armstrong (1982) to suggest an age of 208 Ma for the Triassic–Jurassic boundary.

Armstrong and Besancon (1970) noted a similar discrepancy between eastern and western North America in earlier dating of the Late Triassic – Early Jurassic. They suggested that the discrepancy was due to zeolite-facies burial metamorphism of the basaltic rocks from eastern North America having caused argon loss. However, as argued by Seidemann *et al.* (1984) and Seidemann (1988), argon loss does not seem to have been significant in the more recent radiometric dating of Hettangian flows and intrusions of the Newark Supergroup (Table 2). For example, the $^{40}Ar/^{39}Ar$ release spectrum for the Talcott Basalt (Seidemann *et al.* 1984) shows only enough argon loss to have lowered a conventional K–Ar date by about 3 Ma. The good fit of the K–Ar data from the North Mountain Basalt to an isochron yielding an age of 191 ± 5 (2σ) Ma with an initial $^{40}Ar/^{39}Ar$ ratio less than that of atmospheric argon also suggests that there was little argon loss (Hayatsu 1979). Nor would significant argon loss be expected since the colour of organic tissue in the Newark Supergroup sediments of the Newark, Hartford, and Fundy basins indicates a maximum temperature of only 80–110°C (Robbins 1983). Also, Seidemann (1988) showed that hydrothermal alteration of the Hampden Basalt of the Hartford Basin has more often led to addition of excess argon rather than to argon loss.

Nor does it seem likely that the discrepancy between dating of the Triassic–Jurassic boundary in British Columbia and in the Newark Supergroup is due to problems in paleontological correlation. The correlations in British Columbia should be reliable, being based mainly on ammonites in marine sediments. The paleontological correlations in eastern North America also seem reliable despite being based on continental sediments. Within 30 m of the base of the North Mountain Basalt (Olsen *et al.* 1987) and a correlative of the Orange Mountain Basalt (Cornet 1977, pp. 141, 382), a palynofloral zone with pollen species diagnostic of the Rhaetic is abruptly succeeded by a palynofloral zone strongly dominated by *Corollina meyeriana* and containing species supporting an Early Jurassic age. No unconformity is observed in these 30 m below the basalts, making it very likely that at least the lower part of the *Corollina meyeriana* palynofloral zone and the North Mountain and Orange Mountain basalts are Hettangian. This age assignment is supported by the fossil assemblages in sediments directly overlying the North Mountain Basalt, since the *dominant* Late Triassic reptiles and amphibians are absent from these sediments (despite the sediments being highly fossiliferous and from four different facies), whereas all families present in these sediments were already in existence in the latest Triassic. This strongly supports a sudden and dramatic extinction event at the Triassic–Jurassic boundary, with the North Mountain Basalt and its associated sediments being emplaced soon after in the Hettangian—too soon for originations (Olsen *et al.* 1987).

Most of the other basalt flows of the Newark Supergroup were also assigned paleontologically to the Hettangian by Olsen *et al.* (1982), as shown in Fig. 1. The flows have been correlated between basins using palynomorphs, fossil fish, and reptile footprints. Their assignment to the Hettangian is partly based upon them being within the *Corollina meyeriana* palynofloral zone just above a Rhaetic palynofloral zone and well below palynomorphs characteristic of the late Pliensbachian – Toarcian in the overlying palynofloral zone (Cornet 1977, p. 253). Also, Olsen *et al.* (1987) used Milankovitch lacustrine cycles to argue that the North Mountain Basalt is less than 200 000 years younger than the Triassic–Jurassic boundary and that the Newark Supergroup flows were emplaced within as little as 600 000 years of each other. Further support for a Hettangian age comes from all the flows being of normal polarity, which seems characteristic of the Hettangian, whereas the Sinemurian to Bathonian are characterized by mixed polarity (Kent and Gradstein 1985).

Is it then possible that the age of the Triassic–Jurassic boundary is overestimated by about 15 Ma in recent time scales? In our opinion, yes; however, further radiometric dating will be required to test this. With one exception, the date of 190 ± 3 Ma for the Hettangian of the Newark Supergroup is not in direct conflict with any of the Early Jurassic – Late Triassic dates from British Columbia and elsewhere summarized by Odin and Létolle (1982) in their Fig. 3 and by Kennedy and Odin (1982) in their Table 9 and used by them to place the Triassic–Jurassic boundary at 204 ± 4 Ma. Nor is there direct conflict with the recent dating of the Early Jurassic by Hess *et al.* (1987). The one direct conflict is with two

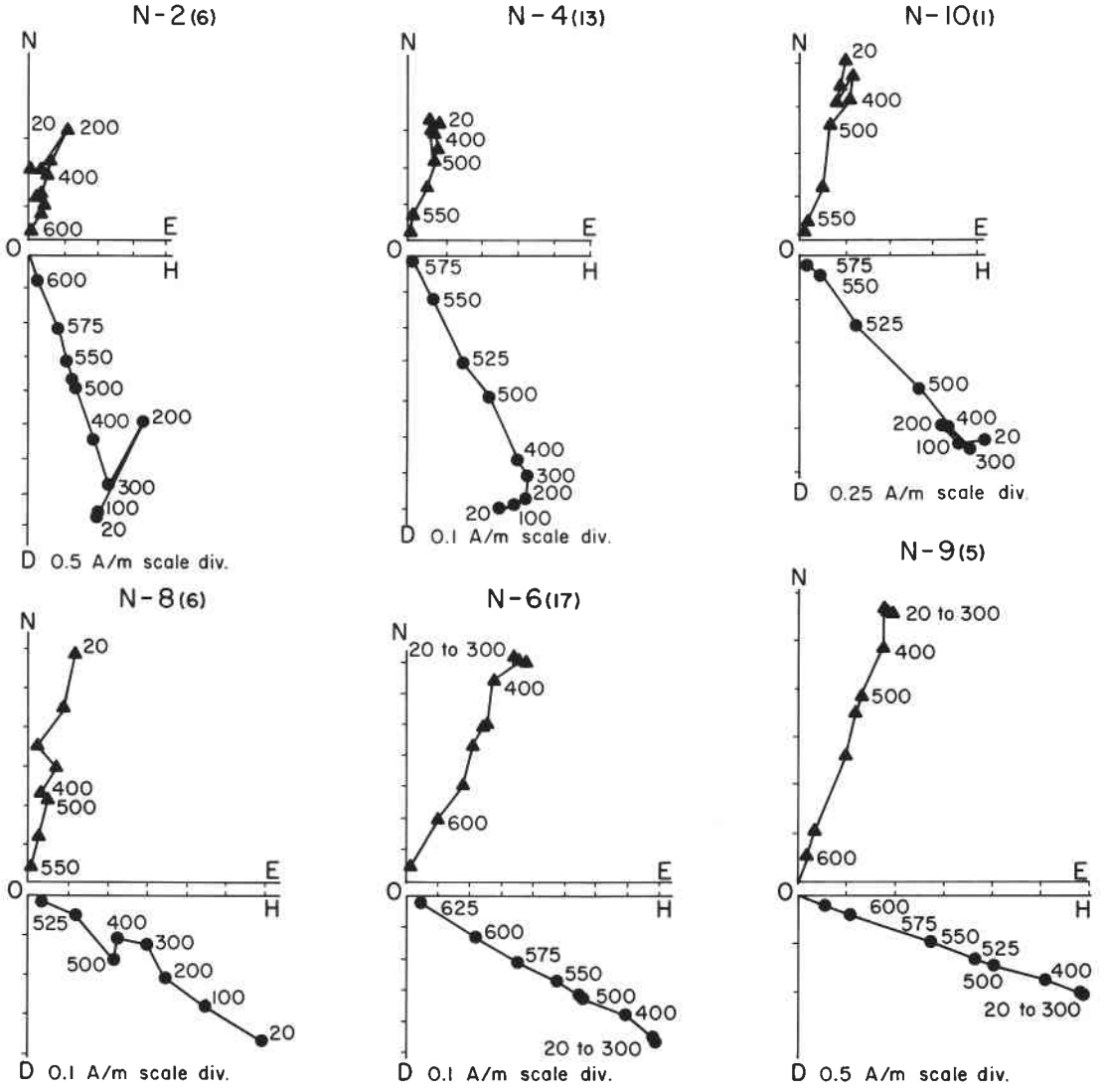


FIG. 5. Representative vector plots for the North Mountain Basalt of the north shore of the Bay of Fundy. In the upper part of each vector plot, each point shows the northward component of remanent magnetization plotted against the eastward component for a given temperature (°C) of thermal demagnetization, successive points being joined by straight lines. A line drawn from a point to the origin will give the declination of remanence observed at that temperature. In the lower part of each vector plot, each point shows the horizontal component of remanence plotted against the downward component. A line drawn from a point to the origin will give the inclination of remanence observed at the given temperature, and the line's length will give the intensity of remanence. All axes for a given sample are drawn to the same scale, and the size of the unit of scale division is given.

K–Ar hornblende dates averaging 204 ± 10 Ma (Wanless *et al.* 1974) from the Topley Intrusions of British Columbia, which are likely late Sinemurian. That is, all the above dates except those from the Topley Intrusions are compatible with a Triassic–Jurassic boundary at 193 ± 6 Ma. Moving the boundary back to 193 Ma will make Early Jurassic ammonite zones of much shorter average duration than Late Triassic ammonite zones (0.6 Ma versus 2.1 Ma using the Aalenian-Bajocian date of Kennedy and Odin (1982) and the Anisian–Ladinian date of Odin and Létolle (1982)); this may be viewed as a drawback. However, a similar difference is required between the average duration of ammonite zones in the latest Jurassic (Tithonian zones of chart 2.10 of Harland *et al.* 1982) and the earliest Cretaceous if the Hawaiian M-sequence magnetic lineations were produced with a constant spreading rate (Fig. 3 of Kent and Gradstein 1985).

In conclusion, we suggest that the Triassic–Jurassic boundary lies at 193 ± 6 Ma, but we recommend further radiometric dating to resolve the discrepancy in dates between British Columbia (the Topley Intrusions in particular) and the Newark Supergroup.

Possible trans-Atlantic correlatives

As shown above, the Hettangian seems to have been a time of major intrusion and extrusion of basaltic magma in eastern North America, producing mostly high-TiO_2 quartz-normative tholeiites with radiometric dates of about 190 Ma. These were likely associated with the early stages of Atlantic rifting and could have trans-Atlantic correlatives (Fig. 1).

Possible northwest African correlatives are high-TiO_2 quartz-normative tholeiitic flows of the Moraccan High Atlas, yielding K–Ar dates of 200 ± 15 Ma (Manspeizer *et al.*

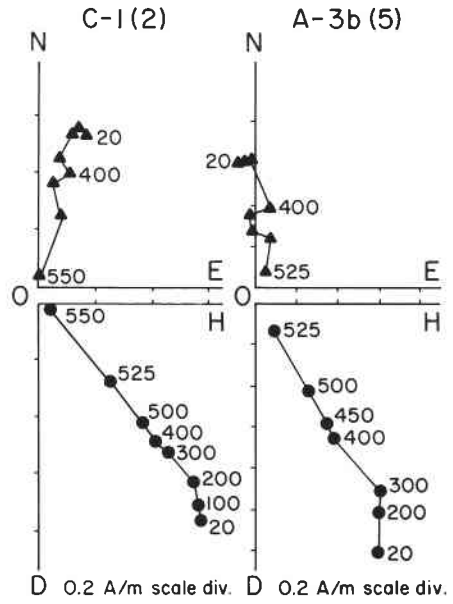


FIG. 6. Representative vector plots of remanent magnetizations observed during thermal demagnetization of the Caraquet and Avalon dike dolerites. See the caption in Fig. 5 for explanation.

1978), and the Foum Zguid dolerite dike of Morocco, yielding K–Ar dates averaging 189 ± 5 Ma (Hailwood and Mitchell 1971).

A possible Iberian correlative is the Messajana dike, which parallels the Avalon dike in predrift configuration. It is likely a high-TiO_2 quartz-normative tholeiite, judging by the data of Alibert (1985), and yields 15 plagioclase and pyroxene K–Ar dates averaging 177 ± 10 Ma (Schott *et al.* 1981).

Paleomagnetic measurements

Pairs of companion specimens (cylinders of ~2.3 cm diameter and ~2.1 cm height) were cut from each block sample collected from the North Mountain Basalt, the Caraquet dike, and the Avalon dike. One specimen of each pair was subjected to stepwise thermal demagnetization (usually to temperatures of 100, 200, 300, 400, 500, 550, 600, 620, 650, and 680°C) using a Schonstedt TSD-1 demagnetizer. The other specimen of each pair was subjected to stepwise alternating-field (AF) demagnetization (usually to fields of 5, 10, 15, 20, 25, 30, 40, 50, 60, and 80 mT) using a Schonstedt GSD-1 demagnetizer. The natural remanent magnetization (NRM) of each sample was measured initially and then after each step in demagnetization to progressively higher fields or temperatures using a Schonstedt SSM-1 magnetometer. Vector plots (Figs. 5, 6) were used for each specimen to help determine its characteristic remanence direction.

Upon stepwise thermal demagnetization, most of the 65 specimens measured reveal a single stable remanence direction as shown in vector plots by data points arranged with little scatter about a straight line through the origin. A small component of magnetization differing in direction from that of the stable component is often seen in vector plots (see N-2 6, N-4 13, and N-10 1 in Fig. 5), but this small component is erased by 400°C. Sixteen specimens show considerable scatter in vector plots (of about the amount shown by N-8 6 of Fig. 5 or somewhat greater) but are still reasonably stable and were used. The stable remanence directions (omitting data from below 400°C) were averaged to give the site means shown in Table 4. The average range of temperatures above 400°C over which the stable remanence direction was observed and averaged for each specimen at a given site is also shown. Seven specimens (N-4 14, N-5 17, N-7 10, N-7 11, N-7 13, N-10 4, C-2 1) were rejected as too unstable upon demagnetization, and one specimen (N-1 4) was rejected because of probable orientation error.

Under stepwise AF demagnetization, the specimens behaved similarly to their companion specimens under thermal demagnetization except that there was usually less scatter in the vector plots. A small component of magnetization differing in direction from that of the stable component was again often seen in vector plots but was erased by about 10–20 mT of AF demagnetization. The stable remanence directions were averaged to give the site means shown in Table 4, which also lists the average range of alternating field over which the stable remanence direction was observed and averaged for samples at a given site.

As can be seen in Table 4, there is no significant difference between the site-mean stable remanence directions yielded by thermal demagnetization and those yielded by AF demagnetization. However, we chose to use only the thermally demagnetized results stable above 400°C in calculating paleopoles to avoid components of remanence reset by burial heating (as discussed later).

Magnetic mineralogy

We would expect the lavas and dikes of Table 4 to have crystallized with titanomagnetite grains of composition $X\mathrm{Fe_2TiO_4}\cdot(1-X)\mathrm{Fe_3O_4}$ with X near 0.6 (Curie point near 135°C), which oxidized on initial cooling producing magnetite grains with fine ilmenite exsolution lamellae. This expectation was shown to be justified for the Avalon and Shelburne dikes (Papezik and Hodych 1980) and is now tested for our North Mountain Basalt samples.

Polished thin sections were examined under the reflected-light microscope for one sample per site of the North Mountain Basalt (the samples of Table 5). All but two of these 10 samples contain magnetite grains of ~0.01 mm diameter, with fine ilmenite lamellae visible (and often with coarse intergrowths as well). Samples N-2 6 and N-7 12 contain only very fine magnetite grains ~0.1 mm or less in diameter, with fine ilmenite lamellae being visible in N-2 6 but not in N-7 12.

The iron–titanium oxide grains in six samples were analyzed with an electron probe microanalyzer (JEOL JXA-50A) calibrated with an Apollo glass standard. Grains with only fine magnetite–ilmenite intergrowths were analyzed so that the electron microprobe (averaging over ~1 μm^2 of surface area) could estimate the composition of the original titanomagnetite grains. The estimates of average X for the samples vary from 0.8 to 0.4 and are listed in Table 5 along with estimates of the original mole fraction Y of hematite in ilmenite grains. For samples N-8 9 and N-10 1, X and Y are accurate enough to yield (Buddington and Lindsley 1964) estimates of the temperature of iron–titanium oxide crystallization of 740 ± 60 and 830 ± 90°C, respectively, and oxygen fugacities given by $-\log_{10} fO_2$ of 17 ± 3 and 15 ± 3, respectively.

Thermomagnetic curves were measured in air for 19 whole-rock specimens. One Curie point per site is listed in Table 5. Except for two of the specimens at site N-3, all the thermomagnetic curves show that magnetite is the dominant magnetic mineral. The Curie point of the magnetite ranges between

TABLE 4. Tilt-corrected site means for stable remanence directions observed upon demagnetization

	Alternating-field demagnetization					Thermal demagnetization				
Site	N	AF range of stability (mT)	D (°)	I (°)	α_{95} (°)	N	Stable from 400°C to (°C)	D (°)	I (°)	α_{95} (°)
N-1	3	10–60	354.6	59.4	10.6	3	550	357.8	60.7	6.7
N-2	4	15–50	11.3	64.0	7.0	4	600	17.7	62.6	9.1
N-3	4	15–40	23.1	53.3	17.7	4	525	9.4	50.4	14.8
N-4	4	10–40	27.2	56.0	8.8	3	555	22.4	57.4	7.3
N-5	4	20–50	34.9	42.4	6.8	3	525	36.8	44.7	6.7
N-6	4	15–70	26.8	28.6	7.6	4	600	25.7	28.5	7.2
N-7	4	7.5–20	36.5	53.5	17.4		Unstable			
N-8	4	10–25	19.6	28.9	11.0	4	550	12.2	25.8	20.2
N-9	5	10–60	14.0	26.2	10.7	5	600	14.5	25.0	10.8
N-10	5	10–25	21.3	52.0	8.0	4	525	20.3	46.6	13.7
C-1	6	20–40	11.2	51.8	1.9	6	550	10.1	47.8	5.1
C-2	6	10–25	21.8	51.5	1.9	5	475	9.1	56.7	13.3
A-3a	5	20–35	10.1	54.5	3.7	5	525	24.4	57.1	9.3
A-3b	6	20–40	2.1	51.3	5.1	6	525	15.4	56.1	6.8

NOTES: N is the number of samples averaged; D and I are the declination and inclination of the cleaned site-mean remanence direction; α_{95} is the half angle of the cone of 95% confidence about the mean.

specimens from the 575°C expected of pure magnetite to as low as 510°C, presumably because of the presence of some titanium in the magnetite. However, a Curie point of 610°C observed for specimens N-2 6 (Fig. 7) and N-6 17 is due to magnetite(?) rather than titaniferous hematite, judging by the high saturation magnetization and NRM observed, but the cause of the magnetite's high Curie point is not known. For specimen N-2 6, the presence of hematite of Curie point ~660°C is also apparent from the thermomagnetic curve and from the red colour of all the samples at this site.

Most of the thermomagnetic curves are reasonably reversible apart from a small reduction in magnetization observed upon cooling and presumably due to some magnetite oxidizing to hematite (as seen to greater than average degree in specimen N-9 3, Fig. 7). None of the thermomagnetic curves show any clear sign of unexsolved titanomagnetite. Evidence of maghemite is only clearly seen in samples N-3 11 (Fig. 7) and N-3 12; their maghemite(?) of Curie point ~420°C transforms at ~450°C into magnetite of Curie point ~530°C. The other samples at site N-3 do not show the presence of maghemite, except possibly for sample N-8 9 (Fig. 7). Two of the specimens (N-10 1 and A-3a 5) show a Curie point of ~580°C if heated crushed but ~530°C if heated in large pieces, perhaps because reducing conditions in the latter case cause titanium to diffuse into the magnetite.

The Curie points for most specimens are consistent with the dominant magnetic mineral in the thermomagnetic curves also being the carrier of the stable remanence. The major exceptions are N-3 11 and N-3 12, which seem to be dominated by maghemite of Curie point ~420°C but whose stable remanence is carried by magnetite with blocking temperatures between 400 and 525°C.

Discussion of the paleomagnetic results

Timing of remanence acquistion

A fold test of when remanence was acquired in our North Mountain Basalt samples was attempted but proved inconclusive. Figure 8 shows the site-mean remanence directions stable above 400°C before and after untilting. There is a small improvement in grouping after untilting, with α_{95} decreasing from 13.3 to 10.7. (Although partial untilting would further decrease α_{95}, this is not a conclusive test of remanence being acquired during tilting because the tilt corrections at sites N-1, N-2, N-3, and N-10 may be significantly overestimated.)

A positive conglomerate test has been reported for the North Mountain Basalt of the south shore of the Bay of Fundy (Carmichael and Palmer 1968); clasts of basalt in the overlying Hettangian sediment have random orientations of NRM. This provides strong evidence that the characteristic remanence direction of the North Mountain Basalt was acquired in the Hettangian during initial cooling of the flows.

Further support for our North Mountain Basalt remanence having been acquired during initial cooling comes from there being a single component of stable remanence carried by magnetite with blocking temperatures mostly between 400 and 580°C. Zeolite-facies alteration typical of the North Mountain Basalt (Aumento 1966) suggests that deep burial has not heated the basalts above 200°C (Turner 1981, p. 420). This maximum temperature follows from the maximum rank of coal commonly associated with zeolite-facies burial metamorphism (Turner 1981, p. 305) and from temperature–time–coal-rank plots such as those of Hood *et al.* (1975). Indeed, colour of organic tissues in Newark Supergroup sediments of the Fundy Basin suggests that heating was to 110°C or less (Robbins 1983). Even heating to 200°C should not have remagnetized magnetite grains with blocking temperatures above 400°C (Pullaiah *et al.* 1975). Hence, with the possible exception of site N-7, the sites were not likely thermally remagnetized by burial heating.

However, it might be argued that burial heating caused chemical rather than thermal remagnetization. Titanomagnetite grains (with $X \sim 0.6$) that escaped oxidation upon initial cooling might have oxidized during subsequent burial heating, producing a chemical remanent magnetization. However, burial heating to 200°C or less would likely cause oxidation to titanomaghemite of Curie temperature 400°C or less (Readman and O'Reilly 1972); this could not carry the remanence that is stable above 400°C. Burial heating to 200°C would not likely

TABLE 5. Site locations and magnetic properties

Site	Site location		Attitude (±5°)		Site-mean remanence intensity (±SD) (A/m)[b]	Site-mean susceptibility (±SD) ($\times 10^{-2}$ SI)[b]	Sectioned sample No.	Curie temp. (±7°C) of sectioned sample (°C)	Composition Fe–Ti oxides	
	Lat. (°N)	Long. (°W)	Strike (°)	Dip (°)					*X* in magnetite grains (±SD)	*Y* in ilmenite grains (±SD)
N-1	45.290	64.775	98	20 S[a]	1.6 (±0.4)	4.2 (±0.8)	2	570	0.40±0.15 (n = 5)	−0.02±0.05 (n = 6)
N-2	45.292	64.773	98	20 S[a]	0.6 (±0.4)	3.3 (±2.9)	6	610		
N-3	45.303	64.767	98	20 S[a]	3.3 (±2.0)	7.1 (±0.1)	8	530		
N-4	45.317	64.757	98	30 S	1.0 (±0.3)	8.8 (±1.2)	16	528	0.47±0.22 (n = 9)	0.09±0.09 (n = 11)
N-5	45.368	64.333	89	20 S	1.2 (±0.5)	2.4 (±0.2)	17	569	0.80±0.03 (n = 5)	
N-6	45.395	64.198	165	24 W	0.7 (±0.3)	0.8 (±0.2)	17	610		
N-7	45.395	64.198	165	24 W	0.6 (±0.2)	5.0 (±1.8)	12	510		
N-8	45.395	64.197	165	24 W	0.9 (±0.4)	6.5 (±2.4)	9	530	0.55±0.08 (n = 4)	0.03±0.01 (n = 5)
N-9	45.385	64.058	300	40 N	4.0 (±1.6)	0.3 (±0.2)	3	580	0.64±0.12 (n = 6)	0.10±0.06 (n = 4)
N-10	45.408	64.003	257	17 N[a]	1.3 (±0.2)	3.6 (±0.6)	1	577	0.44±0.13 (n = 9)	0.04±0.01 (n = 3)
C-1	47.795	64.942	—	—	0.8 (±0.6)	1.1 (±0.2)	2	572		
C-2	47.827	64.888	—	—	1.5 (±0.3)	2.1 (±0.1)	2	530		
A-3a	47.173	52.888	—	—	1.1 (±0.4)	2.9 (±0.3)	5	577		
A-3b	47.175	52.885	—	—	1.5 (±0.3)	2.4 (±0.1)	5	550		

NOTE: SD is standard deviation.
[a] Error in these dips is ±10°.
[b] SI units follow the suggestions of Shive (1986).

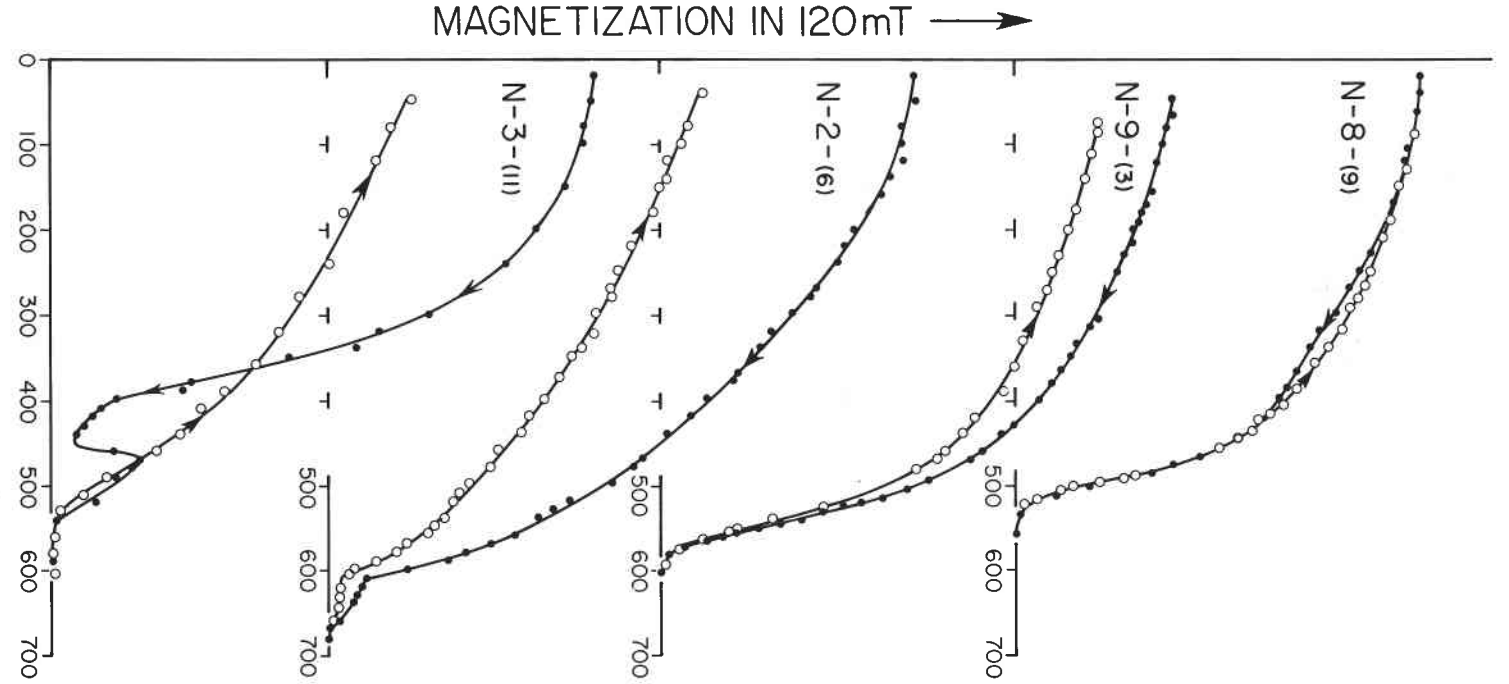


FIG. 7. Examples of thermomagnetic curves observed for the North Mountain Basalt flows of the north shore of the Bay of Fundy.

oxidize titanomagnetite to a magnetite–ilmenite intergrowth (Readman and O'Reilly 1972); but if it did, the process should be very incomplete, and further heating during a thermomagnetic run should convert more titanomagnetite to higher Curie point magnetite. This was not observed; the thermomagnetic curves for most sites from the north shore of the Bay of Fundy

show only magnetite and are reasonably reversible. Hence it is unlikely that these sites were chemically remagnetized by burial heating. Titanomagnetite or titanomaghemite may be more common in basalts from the south shore of the Bay of Fundy (Carmichael and Palmer 1968), but the positive conglomerate test suggests that they too have mostly not been remagnetized.

Seguin *et al.* (1981) suggested that sites along the Caraquet dike at which steeper (~54°) inclinations of remanence are observed (their C and N sites and our C-1 and C-2 sites are in this category) may have remagnetized some 20 Ma or more later than their sites with shallower (~37°) inclinations of remanence, perhaps as a result of heating associated with opening of the Atlantic. This remagnetization hypothesis seems unlikely (except possibly for the N sites of Seguin *et al.*) because there is no evidence of heating to more than ~200°C, whereas a single stable remanence direction carried by magnetite with blocking temperatures mostly above 400°C is characteristic of our sites C-1 and C-2 and probably of the C sites of Seguin *et al.* (1981). Also, reasonably reversible thermomagnetic curves showing magnetite but little or no titanomagnetite or titanomaghemite seem typical of the Caraquet, Avalon, and Shelburne dikes, making it unlikely that such heating would have caused later acquisition of chemical remanence.

The normal polarity of all the Newark Supergroup flows and intrusions assigned paleontologically to the Hettangian and studied paleomagnetically (Table 6) also supports their having magnetized during the Hettangian, since normal polarity seems characteristic of the Hettangian, unlike the mixed polarity of the Sinemurian to Bathonian (Kent and Gradstein 1985).

Paleosecular variation in the Hettangian

Paleosecular variation is best estimated using *between-flow* scatter in "cleaned" remanence directions in a large set of lava flows, correcting for within-flow scatter as done by Merrill and McElhinny (1983). Unfortunately, between-flow scatter has not been determined for any of the sets of Newark Supergroup flows. Although substituting *between-site* scatter for between-flow scatter can underestimate paleosecular variation, it may give a reasonable estimate when the number of sampling sites is not very much larger than the large number of flows sampled, as may be the case with the North Mountain Basalt study of Carmichael and Palmer (1968). Between-site scatter may even give a reasonable estimate of paleosecular variation when there are few flows if these flows are thick enough to have cooled slowly and recorded paleosecular variation, as seems to have occurred in the basal flow of the Preakness Basalt (McIntosh *et al.* 1985).

Table 8 lists paleosecular variation estimated from between-site scatter in remanence directions for all the sets of Hettangian flows of the Newark Supergroup studied paleomagnetically. Site-mean remanence directions (cleaned by AF demagnetization) were averaged for a given set of flows, and the Fisher estimate k of precision was calculated. An estimate of the angular dispersion of the paleofield was then obtained from $S_t = 81/\sqrt{k}$. (The estimate for the North Mountain Basalt of the north shore of the Bay of Fundy may be a little inflated by inaccuracies in tilt correction.) Correction for within-site scatter (following Merrill and McElhinny 1983, p. 202) could be done for all but Carmichael and Palmer's (1968) study but only reduces S_t by a degree or less and has been omitted. Also listed is the present secular variation observed (Irving 1964, p. 146) at the latitude at which the inclination of remanence of

TABLE 6. Virtual paleopoles for basaltic flows and intrusions of the Newark Supergroup paleontologically assigned to the Hettangian

Unit	Number of sites	Demagnetization	Tilt corrected	Virtual paleopole position				Source
				Lat. (°N)	Long. (°E)	dp (°)	dm (°)	
North Mountain Basalt flows (north shore, Bay of Fundy)	9	TH	Yes	66.4	71.9	8.6	13.5	This paper
North Mountain Basalt flows (south shore, Bay of Fundy)	25	AF	Yes	73	104	4.5	6.5	Carmichael and Palmer (1968)
Orange Mountain Basalt flows	5	AF	Yes	65.3	106.2	3.0	5.4	McIntosh *et al.* (1985)[a]
Preakness Basalt flows	9	AF	Yes	61.5	76.4	6.9	12.6	McIntosh *et al.* (1985)[a]
Hook Mountain Basalt flows	4	AF	Yes	63.3	97.1	7.4	13.6	McIntosh *et al.* (1985)[a]
Talcott Basalt flows	10	AF	Yes	51.1	78.3	1.8	3.6	de Boer (1968)[a]
Holyoke Basalt flows	10	AF	Yes	60.1	84.7	2.6	4.7	de Boer (1968)[a]
Palisades sill and outlier	3	AF	Yes	65.8	104.0	7.1	12.7	Opdyke (1961)[a]
Fairhaven dike	7	AF	Yes	66.1	54.0	8.2	13.3	Smith (1976)[a]

NOTES: Lat. and Long. are the latitude and longitude of the virtual paleopole with its oval of 95% confidence having semimajor and semiminor axes dm and dp calculated from the average of site-mean remanence directions cleaned thermally (TH) or by alternating-field (AF) demagnetization.

[a]Recalculated. In the case of Smith (1976) and Smith and Noltimier (1979) only dp and dm were recalculated (from quoted α_{95}).

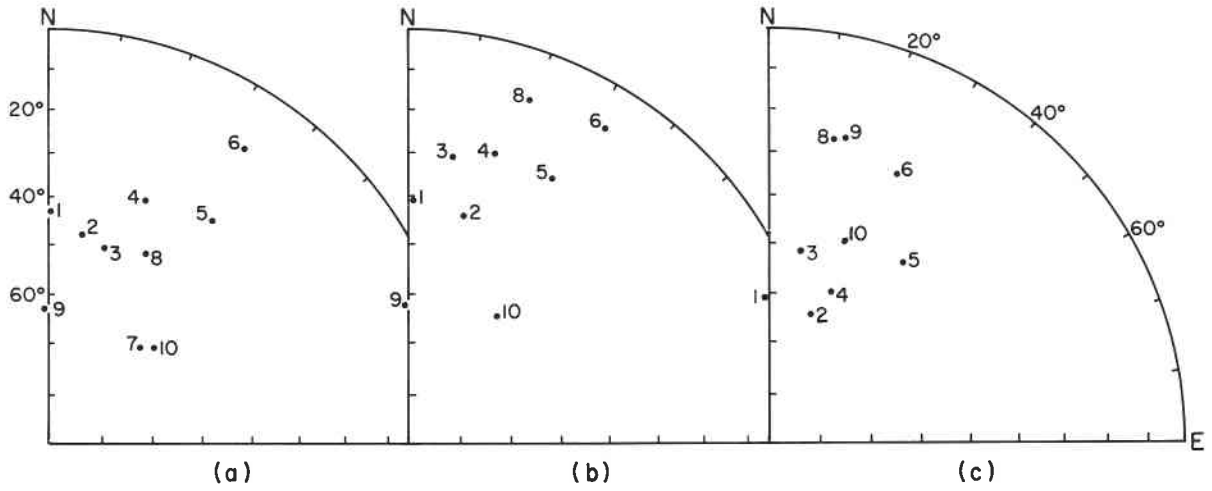


FIG. 8. Stereoplots of site-mean remanence directions of the North Mountain Basalt of the north shore of the Bay of Fundy compared before any demagnetization or tilt correction (*a*), after thermal cleaning but before tilt correction (*b*), and after thermal cleaning and tilt correction (*c*). All remanences observed were of normal polarity.

the given set of flows was acquired.

The results in Table 8 tend to support the conclusions of Irving and Pullaiah (1976), who averaged worldwide data for the Late Triassic – Early Jurassic and concluded that secular variation then and now do not greatly differ.

However, the low paleosecular variation estimates in Table 8 for the Orange Mountain, Holyoke, and Talcott basalts deserve comment. Beck (1972) obtained similar low estimates from the low between-site scatter in remanence directions in basaltic sills of the Gettysburg Basin and suggested that low paleosecular variation was characteristic of the Late Triassic – Early Jurassic. The low between-site scatter might instead be due to much of the paleosecular variation having been averaged out by burial heating causing maghemitization (as suggested by T. E. Smith 1976) or exsolution of magnetite–ilmenite intergrowths from titanomagnetite. However, this hypothesis is not supported by our thermomagnetic curves, which suggest that maghemite or unexsolved titanomagnetite is not common in these lavas. Of the five thermomagnetic curves that we ran from five sites in the Holyoke flows, none shows any sign of titanomagnetite or maghemite; all the curves are reasonably reversible with Curie points near 580°C. Of the nine thermomagnetic curves that we ran for six sites in the Talcott flows, only two curves show the presence of maghemite; the other curves are reasonably reversible with Curie points near 580°C (except for one near 480°C) and with no sign of maghemite or unexsolved titanomagnetite. Perhaps the low paleosecular variation estimates from these flows result from using between-site rather than between-flow scatter and from sampling too few flows.

Comparing Canadian and American Hettangian virtual paleopoles from basaltic rocks

The virtual paleopoles from North American basaltic flows and intrusions assigned paleontologically to the Hettangian are listed in Table 6, and those assigned radiometrically to the Hettangian are listed in Table 7. The North Mountain Basalt (north shore), the Avalon dike (east end), and the Caraquet dike paleopoles are calculated from the thermally cleaned site means of Table 4. In the case of the Caraquet paleopole, the thermally cleaned site means of Seguin *et al.* (1981) are also used (excluding their N sites because they may not be stable above 400°C).

The five virtual paleopoles from Atlantic Canada are on average 11.2° more northerly than the five virtual paleopoles from the flows of the northeastern United States (Fig. 9). However, the difference is not significant at the 95% confidence level (α_{95} being 6.5 and 7.8°, respectively, for the two sets of paleopoles). Also the difference decreases to 9.2° when the Fairhaven and Palisades paleopoles are added and to 6.4° when the remaining paleopoles are also added (with α_{95} decreasing to 7.2 and 4.8°, respectively). This suggests that the difference is due to incomplete averaging out of paleosecular variation. Certainly the North Mountain Basalt is not likely to have adequately averaged out paleosecular variation as argued by Carmichael and Palmer (1968) using the distinct difference in azimuth of remanence between the upper and lower flows of the south shore of the Bay of Fundy. Nor are any of the dikes of Atlantic Canada in Table 7 likely to have cooled long enough to have adequately averaged out paleosecular variation as discussed for the Shelburne dike by Carmichael and Palmer (1968). Even together, these flows and dikes of Atlantic Canada do not seem to have adequately averaged out paleosecular variation. The same seems true of the flows of the United States in Table 6.

Other possible explanations for the difference in Hettangian paleopoles between Atlantic Canada and the northeastern United States seem much less likely. One might try to explain the difference by suggesting that the basaltic rocks of Atlantic Canada are younger than those of the northeastern United States. This would require about a 20 Ma age difference based on the polar wander curve for North America (Irving and Irving 1982), which is unlikely for the flows because they are all paleontologically assigned to the Hettangian. One might attempt to explain the difference by postulating that the flows and dikes of Atlantic Canada have a steep downward remanence component superimposed long after initial cooling and incompletely removed by demagnetization. This is unlikely for the reasons used above to argue that the remanence stable above 400°C was acquired during initial cooling. One might even try to explain the difference by suggesting that Atlantic Canada moved southward relative to cratonic North America

by up to 11° (~1200 km) since the Early Jurassic. This is very unlikely because the North American seaboard from Nova Scotia to Georgia separated from Africa in the Early Jurassic with little or no strike-slip motion, as indicated by the sea-floor magnetic anomaly pattern (Emery and Uchupi 1984, p. 232).

The Hettangian paleopole for North America and critique of the red-bed data

The Newark Supergroup flows and intrusions assigned paleontologically to the Hettangian (Olsen *et al.* 1982) are the most likely to record the Hettangian paleofield of North America. The average of their virtual paleopoles (Table 6) yields a paleopole at 64.5N, 85.4°E with $\alpha_{95} = 6.1°$. (The intrusions of Table 7 are less certainly Hettangian, but their average paleopole at 69.7°N, 85.0°E with $\alpha_{95} = 5.4°$ is not sigificantly different.)

Hettangian red-bed results will be excluded from the paleopole determination for reasons now to be discussed. Consider the Late Triassic – Early Jurassic red beds of the Newark basin studied by McIntosh *et al.* (1985). Although those of the red beds with normal polarity have an average cleaned remanence direction ($D = 7°$, $I = 26°$) very close to that of the intercalated flows ($D = 7°$, $I = 27°$), the red beds with *reverse* polarity have an average cleaned remanence direction ($D = 191°$, $I = -1°$) that is not antiparallel. McIntosh *et al.* attributed this nonantiparallelism to the presence in the reverse red beds of a steeply downward Cenozoic remanence component not completely removed by thermal demagnetization. However, if they were correct, a similar Cenozoic component would be expected in the normal cleaned red beds, since the normal and reverse red beds are not said to differ on average in lithology or magnetic properties. As shown in Fig. 10, we then expect the normal and reverse red-bed remanences to have been antiparallel originally and at ~14° to horizontal before an equal Cenozoic component not removable by demagnetization was added to each, deflecting the normal and reverse cleaned remanences to their present 26° and −1° inclinations, respectively. That is, complete removal of the suspected Cenozoic component should give the Hettangian red beds (normal polarity) an inclination of remanence about 13° shallower than the intercalated Hettangian flows. This shallower inclination in the red beds is suspect because it might be caused by compaction of the red beds, which are said to be clay rich. Although Opdyke (1961) argued against a compaction-induced inclination error by pointing out the similar inclinations of remanence in the red beds and intercalated basalts of the Newark basin, his red beds may have shown a shallower remanence had they been thermally demagnetized.

Compaction-induced inclination error could be tested for by comparing coarser grained red beds (which should be less compacted) with finer grained red beds. Although McIntosh *et al.* (1985) did not mention such a test, the Hettangian (Olsen *et al.* 1982) Turners Falls red sandstones of the Deerfield Basin, which are coarser than the red beds of McIntosh *et al.*, do show a steeper inclination of thermally demagnetized remanence ($D = 10°$, $I = 37°$; Brown 1979). However, this steeper remanence is also suspect because the greater porosity of the Turners Falls red sandstones makes them more susceptible to acquiring remanence during diagenetic changes long after deposition (as documented for the Lower Triassic Moenkopi red sandstones of Colorado by Larson *et al.* 1982).

In conclusion, it is safer to exclude the red-bed data from the Hettangian paleopole determination for North America until their paleomagnetic reliability is further tested.

TABLE 7. Virtual paleopoles for basaltic intrusions possibly related to Hettangian flows of the Newark Supergroup, judging by radiometric dates

Intrusion	Number of sites	Demagnetization	Tilt corrected	Virtual paleopole position				Source
				Lat. (°N)	Long. (°E)	dp (°)	dm (°)	
Shelburne dike	2	AF	No	69	98	3.5	5.5	Larochelle and Wanless (1966)
Avalon dike (east end)	2	TH	No	72.4	64.1	11.5	15.9	This paper
Caraquet dike	8	TH	No	74.9	104.0	7.7	11.3	This paper and Seguin *et al.* (1981)
Mt. Carmel sill	3	AF	Yes	66.7	87.3	3.3	5.8	Smith and Noltimier (1979)[a]
Cross Rock dike	6	AF	Yes	71.3	45.2	8.1	12.0	Smith (1976)[a]
Buttress dike	9	AF	Yes	68.7	46.2	7.6	11.6	Smith (1976)[a]
West Rock sill	11	AF	Yes	63.2	86.0	3.0	5.4	Smith and Noltimier (1979);[a] de Boer (1968)
Higganum dike	11	AF	No	65.5	99.8	2.1	3.7	Smith (1976)[a]
Gettysburg sill	10	AF	Yes	63.2	101.5	3.3	6.1	Smith (1976);[a] Beck (1972)
Frederick dikes	10	AF	Yes	69.0	103.7	2.0	3.4	Smith (1976)[a]

NOTES: See footnotes to Table 6 for explanation of symbols.

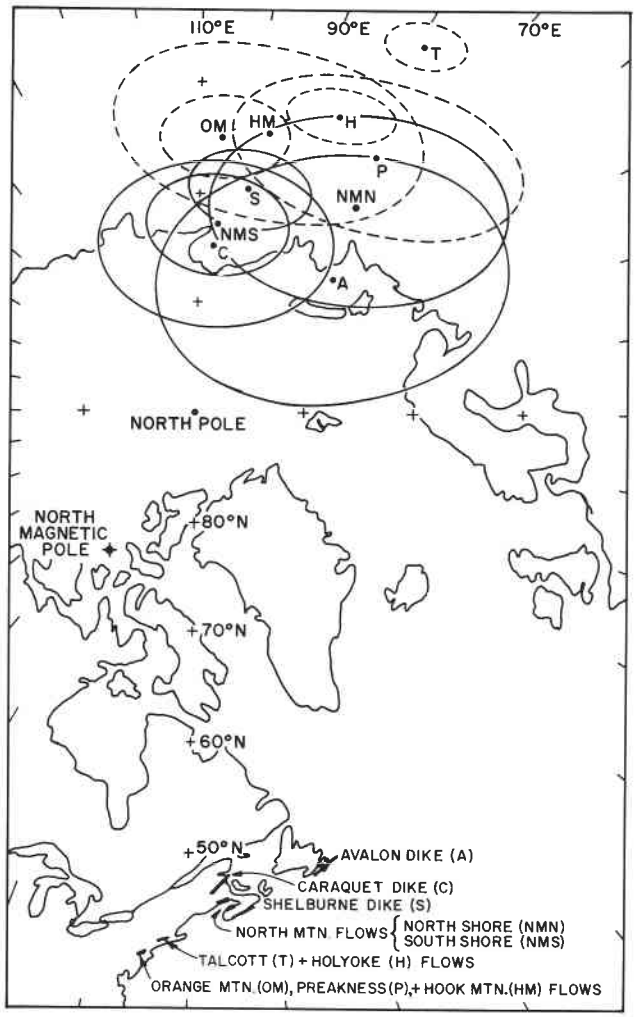


FIG. 9. Virtual paleopoles for the basalt flows of eastern North America paleontologically assigned to the Hettangian and for dolerite dikes of Atlantic Canada radiometrically assigned to the Hettangian. The ovals of 95% confidence for the virtual paleopoles are solid for Atlantic Canada and dotted for the northeastern United States.

Paleomagnetic correlation of Late Triassic – Early Jurassic basaltic rocks of eastern North America

It was found by de Boer (1968) that the stable remanence directions for the basalt flows of the Hartford Basin showed little between-site scatter but had mean inclinations that increased from the oldest to the youngest set of flows—from 12° for the Talcott, to 25° for the Holyoke, to 42° for the Hampden flows. Then de Boer (1968) and de Boer and Snider (1979) attempted correlations between basaltic flows and intrusions of eastern North America, based particularly upon inclinations of remanence. However, this led de Boer (1968) to correlate the North Mountain Basalt with the Hampden flows, whereas Olsen *et al.* (1982) paleontologically correlated the former with the early Hettangian and the latter with the early Sinemurian. We judge the paleontologic correlations to be more reliable because even the flows and intrusions showing little between-site scatter in remanence direction may not have averaged out paleosecular variation adequately.

T. E. Smith and Noltimier (1979), on the other hand, concentrated on the difference in longitude rather than latitude that

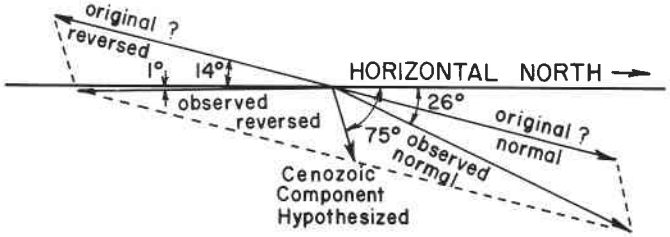


FIG. 10. Demonstration that nonantiparallelism of observed normal and reversed remanences in the red beds of the Newark Basin (McIntosh *et al.* 1985) could be due to a Cenozoic remanence component added equally to normal and reverse remanences that were originally antiparallel and inclined at ~14° to horizontal.

they found between two groups of virtual paleopoles from Late Triassic – Early Jurassic basaltic rocks of the eastern United States. Group I had a mean paleopole at 63.0°N, 83.2°E and consisted of flows and intrusions assigned to the Hettangian–Sinemurian on the basis of paleontology or of radiometric dates of ~195 Ma (recalculated). Group II had a mean paleopole at 65.3°N, 103.2°E and consisted of intrusions assigned to the Toarcian–Bajocian because many yielded radiometric dates of ~179 Ma (recalculated). This led Smith and Noltimier to assign the Shelburne dike to group II because of paleomagnetism, whereas our radiometric dating places it in group I. Even more seriously, Smith and Noltimier's paleomagnetic correlation method would lead one to assign the North Mountain, Orange Mountain, and Hook Mountain basalt flows to group II, whereas paleontology and radiometric dating would assign them to group I. Furthermore, Philpotts and Martello (1986) described field relations showing that the Fairhaven dike of group II fed the Talcott flows of group I. They also presented geochemical data suggesting that the Higganum, Buttress, and Bridgeport dikes of group II fed the Talcott, Holyoke, and Hampden flows, respectively, of group I. We suggest that the group II as well as the group I intrusions are mostly Hettangian–Sinemurian in age, with their difference in longitude of paleopole being a result of incomplete averaging out of paleosecular variation.

In conclusion, paleomagnetic correlation of the Late Triassic – Early Jurassic basaltic rocks of eastern North America seems more difficult than these earlier workers assumed, probably because of incomplete averaging out of paleosecular variation.

The Early Jurassic apparent polar wander path for North America

Irving (1979), Van der Voo (1981), Gordon *et al.* (1984), and May and Butler (1986) have pointed out a bend in the Early Jurassic apparent polar wander path for North America. Such a bend is expected to have accompanied initiation of active sea-floor spreading between North America and Africa, which was estimated to have occurred at about 180 Ma (Pliensbachian) by Emery and Uchupi (1984) using sea-floor magnetic lineations and Van Hinte's (1976) time scale. However, this estimate is not very accurate because of the absence of magnetic lineations in the 400 km wide smooth magnetic zone adjacent to the North American and North African continental slopes.

One might hope to better date the start of Atlantic sea-floor spreading and the accompanying change in North American plate motion (Gordon *et al.* 1984) by accurately defining the

TABLE 8. Estimates of paleosecular variation from Hettangian basalt flows of North America

Formation	Source	Approximate thickness (m)	Number of sites sampled	Number of flows sampled	Paleosecular variation[a] (°)	Present secular variation[b] (°)
North Mountain Basalt (south shore, Bay of Fundy)	Carmichael and Palmer (1968)	300	25	≤16	14	14
North Mountain Basalt (north shore, Bay of Fundy)	This paper	240	10	2–6	16	14
Hook Mountain Basalt	McIntosh *et al.* (1985)	110	4	2	11	14
Preakness Basalt	McIntosh *et al.* (1985)	300	9	2 or 3	18	16
Orange Mountain Basalt	McIntosh *et al.* (1985)	130	5	2 or 3	5	16
Holyoke Basalt	de Boer (1968)	130	10	≥2	7	16
Talcott Basalt	de Boer (1968)	70	10	2	6	17

[a]Paleosecular variation estimated from the between-site scatter of AF cleaned remanence directions using the estimate of angular dispersion $S_t = 81/\sqrt{k}$, where k is the Fisher estimate of the precision of the average of site-mean remanence directions. Note that this use of between-*site* rather than between-*flow* scatter may underestimate paleosecular variation.

[b]Approximate present secular variation at latitude at which remanence was acquired.

expected bend in the apparent polar wander path for North America. However, this may prove difficult. Red beds of the Newark Supergroup have the required stratigraphic range, but their paleomagnetic reliability needs further testing. The basalt flows of the Newark Supergroup should be paleomagnetically reliable but seem confined to the Hettangian and lower Sinemurian. The flows of the western Cordillera have a wider stratigraphic range but seem mainly to be in suspect displaced terranes (Rees *et al.* 1985). Use of the dikes and sills along the Atlantic seaboard requires resolving the discrepancy in radiometric dating between Late Triassic – Early Jurassic igneous rocks of eastern North America and those of British Columbia.

For example, May and Butler (1986) considered only T. E. Smith and Noltimier's (1979) group I and group II paleopoles and six other paleopoles as reliable for defining the Jurassic apparent polar wander path for North America. They used Sutter and Smith's (1979) radiometric dating of the intrusions in groups I and II and the time scale of Harland *et al.* (1982) to assign the group I paleopole (~195 Ma) to the Pliensbachian and the group II paleopole (~179 Ma) to the Bathonian–Bajocian. We have presented evidence that group I and group II paleopoles are both likely to be Hettangian–Sinemurian.

Conclusions

(1) The K–Ar isochrons for the Shelburne, Avalon, and Caraquet dikes of Atlantic Canada yield dates of 193 ± 2, 189 ± 3, and 191 ± 2 Ma, respectively, which do not differ significantly from radiometric dates (Table 2) for the North Mountain Basalt flows of Nova Scotia or other flows of the Newark Supergroup assigned paleontologically (Olsen *et al.* 1982, 1987) to the earliest Jurassic (Hettangian).

(2) Previously published geochemistry also suggests that the Shelburne, Avalon, and Caraquet dikes may be related to the North Mountain Basalt and to the Hettangian Orange Mountain and Talcott flows of the Newark and Hartford basins, since all these dikes and flows fall within or near to the high-TiO_2 quartz-normative tholeiite field of Weigand and Ragland (1970).

(3) The Shelburne, Avalon, and Caraquet dikes and the North Mountain flows are thus probably related and were likely emplaced as Hettangian precursors to active Atlantic sea-floor spreading. Their trans-Atlantic correlatives may be the Foum Zguid dike of Morocco, the Messajana dike of Iberia, and some of the flows of the High Atlas (Fig. 1).

(4) The Hettangian flows of eastern North America suggest a date of 193 ± 6 Ma for the Triassic–Jurassic boundary, whereas the Late Triassic – Early Jurassic flows and intrusions of British Columbia (Armstrong 1982) suggest a date some 15 Ma older. Further work is recommended to resolve this discrepancy.

(5) The North Mountain Basalt samples from nine sites on the north shore of the Bay of Fundy are shown to carry a remanence that is stable above 400°C and was likely acquired during initial cooling in the Hettangian. This stable remanence yields a virtual paleopole at 66°N, 72°E (dp = 9°, dm = 14°). Similar results were obtained from two Avalon and two Caraquet dike sites radiometrically assigned to the Hettangian (Tables 6 and 7).

(6) That secular variation in the Hettangian was similar to that at present tends to be supported by estimates from between-site scatter of remanence directions in sets of North American Hettangian flows (Table 8).

(7) The Hettingian flows and dikes of Atlantic Canada yield virtual paleopoles lying on average about 11° farther north than the average for the Hettangian flows of the northeastern United States (Fig. 9). This discrepancy is likely a fortuitous result of incomplete averaging out of paleosecular variation.

(8) The Hettangian paleopole for North America is estimated to lie at 65°N, 85°E (α_{95} = 6°), averaging the virtual paleopoles from the sets of flows and intrusions of Table 6.

(9) Attempts at paleomagnetically correlating Late Triassic – Early Jurassic flows and intrusions of eastern North America (de Boer 1968; T. E. Smith and Noltimier 1979) have not been very successful because of insufficient averaging out of paleosecular variation in these units and because of only normal polarity when they were emplaced.

(10) The bend in the Early Jurassic apparent polar wander curve for North America expected to accompany initiation of Atlantic sea-floor spreading may be difficult to define precisely for the following reasons. (*i*) The paleomagnetic reliability of the red beds requires further testing; for example, compaction of the clay-rich red beds of the Newark Basin may have caused their inclinations of remanence to be reduced by ~13°. (*ii*) Most flows from cratonic North America are confined to the Hettangian or Sinemurian. (*iii*) Intrusions may be of wider age range, but their precise ages will be in doubt until the

above-mentioned discrepancy in Late Triassic – Early Jurassic radiometric dating is resolved.

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